MIL-P-45805C(AL)
4 December 1985
SUPERSEDING
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14 September 1979

MILITARY SPECIFICATION

POLISHING MACHINE, AUTOMOTIVE CRANKSHAFT AND CAM SHAFT

This specification is approved for use within the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of motor driven polishing machine of the abrasive belt type used primarily to produce a smooth finish on automotive crankshaft bearing journals, cam shaft bearing journals, and cams.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152	- Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The polishing machine shall be new and of the manufacturer's latest design which shall meet the performance requirements specified herein. The polishing machine shall impart a smooth finish to the bearing journals of automotive crankshafts and other rotating parts that can be mounted between centers. The design shall be such that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the machine or any machine component.

* 3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the polishing machine. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

* 3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

* 3.2.3 Maintainability. Design and configuration of the polishing machine shall assure that maintenance of operational capability and accuracy shall be reduced to the simplest form of procedures, requiring a minimum of skills and common tools. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

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* 3.2.4 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type of model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.5 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the polishing machine to remove the temporary caps or stoppers before start-up or use of the machine.

3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The polishing machine shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The polishing machine shall be constructed to be capable of performing the operations specified herein. The machine and its equipment shall be complete so that, when installed and connected to a proper power source, it can be used for all operations for which it was designed.

* 3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

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* 3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.4 Surfaces. Castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.5 Performance and product characteristics.

3.5.1 Capacity. The polishing machine shall be capable of handling any crankshaft up to, and including, 60 inches in length with bearing journals up to, and including, five inches in diameter with a stroke of at least seven inches.

3.5.2 Polishing efficiency. The efficiency of the polishing machine shall be such that it shall be capable of performance in accordance with the following, while polishing bearing journals of the largest crankshaft within the capacity of the machine.

(a) Reduce the surface roughness of ground bearing journals from 16 to 20 microinches, arithmetical average (aa), to 4 to 8 microinches, aa.
(b) Accomplish the operation described in (a) above at a rate of at least one bearing journal every 30 seconds.

3.6 Components. The polishing machine shall primarily consist of a floor mounting, ways, headstock and tailstock assemblies, rocker arm assembly, drive belts, abrasive belts, motors and motor controls, and such other components necessary to produce a machine capable of meeting the requirements of this specification.

3.6.1 Floor mounting. The floor mounting shall be fabricated of cast iron, steel, or any combination thereof. The strength of the mounting shall be such that it shall be capable of supporting all components of the machine, plus the heaviest crankshaft within the capacity of the polishing machine, without physical damage or measurable deflection.

3.6.1.1 Splash pan. A splash pan shall be included in the floor mounting to trap surplus polishing compound and abrasive dust. The splash pan shall extend the full length of the polishing machine. The pan shall be accessible for cleaning and shall be provided with an outlet for drainage.

* 3.6.2 Ways. The ways shall be tubular steel or hardened and ground solid rods with ball bushings. The length and alignment of the ways shall be such as to meet the capacity and performance requirements specified herein. Two ways shall be provided for the rocker arm assembly and two for the tailstock assembly.

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3.6.3 Headstock assembly. The housing of the headstock assembly shall be an integral part of the polishing machine base or mounting. The headstock spindle shall turn in lubricated-for-life type ball bearings. A lock nut, or similar means, shall be provided for control of spindle end play.

3.6.3.1 Headstock center. The headstock center shall be made of tool steel and shall be retained in the spindle by means of a set screw, draw bolt, or slow taper.

3.6.3.2 Drive plate. A drive plate, capable of turning crankshafts of any size and weight within the capacity of the polishing machine, shall be mounted on the headstock spindle. The face of the drive plate shall be made of non-metallic material and shall be held in contact with the flywheel flange of the driven crankshaft by springs under compression.

3.6.3.3 Work drive. Power shall be transmitted from the work drive motor to the headstock spindle through V-belts and countershaft with pulleys designed to produce a headstock spindle speed of 46 to 54 revolutions per minute (rpm).

3.6.4 Tailstock assembly. The housing of the tailstock assembly shall be an iron casting and shall be mounted on the ways to provide adjustability of the distance between centers, from zero to 60 inches, with the center fully extended. The assembly shall incorporate a mechanical means of clamping the tailstock at any point within the range of adjustment.

3.6.4.1 Pneumatic cylinder assembly. The tailstock shall incorporate a cylinder assembly designed to provide pneumatically powered center extension. The piston shall be keyed to the cylinder to prevent rotation, and shall be bored to receive the tailstock center.

* 3.6.4.2 Tailstock center. The tailstock center shall be of the live (rotating) type using ball thrust bearings, sealed to prevent entry of foreign materials.

3.6.4.3 Operating air pressure. The cylinder assembly shall provide sufficient thrust to overcome the resistance offered by the drive plate springs (see 3.6.3.2), and to clamp and hold, between centers, crankshafts of any size within the capacity of the polishing machine. The cylinder assembly shall perform satisfactorily when supplied with compressed air at a maintained pressure of 75 to 80 pounds per square inch gage (psig).

3.6.4.4 Air control valve. The air control valve shall be foot operated and shall be furnished with at least 15 feet of, 1/4 inch minimum inside diameter, oil-resistant rubber hose with connections from the valve to the hose and from the hose to the air inlet of the cylinder assembly. The hose shall be capable of withstanding air pressure at least twice as great as the rated operating pressure of the cylinder assembly.

* 3.6.5 Rocker arm assembly. The rocker arm assembly shall be mounted on a cast iron base or a fabricated steel assembly provided with ball bearings at the points of contact with the ways. The base shall be assembled to the ways to provide the longitudinal travel necessary to meet the capacity and performance requirements of this specification. The top of the base shall be machined to the extent necessary to provide a mounting pad for the abrasive belt drive motor.

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3.6.5.1 Rocker arm. The rocker arm shall be mounted to enable the abrasive belt to follow the motion of the rotating crankshaft throw bearing journals, and remain in contact with the bearing journals throughout the complete cycle of rotation. The mounting of the rocker arm shall also permit at least one inch of lateral movement of the abrasive belt, at the point of contact, with a line bearing journal three inches or more in diameter. A counterweight shall be provided to return the rocker arm to a vertical position when not in use. A handle shall be provided at the extreme end of the rocker arm to afford control during operation.

3.6.5.2 Abrasive belt pulleys. The abrasive belt drive pulley shall be mounted on the motor shaft and the idler pulley shall be mounted at the opposite extremity of the rocker arm. The pulleys shall be made of cast iron or aluminum. The mounting of the idler pulley shall be spring loaded to provide continuous abrasive belt tension during operation, and to allow slackening of the tension to the extent necessary to facilitate belt changing. The idler pulley shall turn on bronze sleeve bearings or sealed ball bearings, and shall be hooded to prevent accidental contact with the abrasive belt. Both pulleys shall be crowned to provide correct traction of abrasive belts of any width up to 1 3/4 inches.

3.6.5.3 Abrasive belts. The abrasive belts shall be of the continuous, cloth-backed type with the correct direction of rotation indicated on each belt. The abrasive agent shall be aluminum oxide. The grit size and the bonding agent shall be such that each belt shall be capable of reducing the surface roughness, as specified in 3.5.2, of all bearing journals on crankshafts of any size within the capacity of the polishing machine. Abrasive belts shall be furnished in four different widths - one inch, 1 1/4 inches, 1 1/2 inches, and 1 3/4 inches (see 3.8).

* 3.6.6 Electrical components. Electrical components, including relays, starters, switches, wiring, and their components, shall conform to, and be located in accordance with, NEMA ICS 1 and NFPA 79, as applicable.

* 3.6.6.1 Motors. The polishing machine shall be equipped with two electric motors conforming to CC-M-1807. The motors shall meet the requirements for a totally enclosed, fan cooled, induction squirrel cage type, and shall be rated for continuous duty. Unless otherwise specified (see 6.2.1), the motors shall be dual voltage 230/460 volts alternating current (ac), three phase, 60 hertz (Hz), and be initially wired for operation on 230 volts. Minimum horsepower (hp) ratings shall be two hp for the abrasive belt drive motor, and 1/3 hp for the work drive motor.

3.6.6.2 Junction box. All motor and control wiring shall terminate in a junction box or boxes, located on the back of the machine, to facilitate permanent installation.

3.6.6.3 Motor switches. Each motor shall be controlled by an individual, toggle or push button type, on-off switch located on the front of the machine at the headstock end. The switches shall afford overload protection for the motors.

* 3.6.6.4 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

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3.7 Dielectric strength. The polishing machine electrical system shall withstand, without damage, a root mean square (rms) voltage of 1000 plus twice its rated voltage applied for a period of one minute between mutually insulated circuits, and between insulated circuits and ground. The voltage shall be at a frequency of 60 Hz.

3.8 Additional abrasive belts. Unless otherwise specified (see 6.2.1), twenty four abrasive belts of each width shall be furnished with each machine. The belts shall conform to the requirements specified in 3.6.5.3.

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The polishing machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.11 Lubrication plate or chart. A corrosion-resistant metallic lubrication plate, with permanently legible characters in the English language, shall be permanently attached to each machine in a location visible to the operator. If a chart is furnished, it shall be sealed in a transparent plastic folder or laminated between permanently sealed clear plastic sheets with suitable means for mounting. The following information shall be furnished on the plate or chart:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of lubricant, if applicable.

* 3.12 Workmanship. Standards of workmanship shall assure that the polishing machine shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The

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absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Functional test (see 4.6.2).
(c) Performance test (see 4.6.3).
(d) Dielectric strength test (see 4.6.5).
(e) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be as specified in MIL-STD-105 using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	1.0
Functional test (4.6.2)	S-4	1.0
Performance test (4.6.3)	S-4	1.0
Dielectric strength test (4.6.5)	S-4	1.0
Packaging inspection (4.7)	S-4	4.0

* 4.5 Product examination. Visually and manually examine the polishing machine to determine conformance with the requirements of 3.2 and 3.3, 3.4 through 3.4.4, 3.6 through 3.6.6.4, and 3.8 through 3.12. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the polishing machine has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

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4.6 Tests.

4.6.1 Test conditions. The tests of 4.6.2 through 4.6.5 shall be conducted while the polishing machine is being operated in accordance with the handbook of instructions furnished by the manufacturer (see 6.3).

4.6.2 Functional test. A length of center drilled round steel bar stock, three inches nominal diameter by one to two feet in length, shall be mounted between the centers of the polishing machine. The machine shall then be operated to the extent necessary to test the following features:

(a) Headstock spindle rpm (see 3.6.3.3).
(b) Operation of tailstock assembly (see 3.6.4 to 3.6.4.4).
(c) Operation of rocker arm assembly (see 3.6.5.1).
(d) Abrasive belt traction (see 3.6.5.2).

Failure of any component or assembly to perform its intended function in accordance with the requirements shall be cause for rejection of the machine.

4.6.3 Performance test. The performance tests shall be performed using one crankshaft and one cam shaft taken from a four, six, or eight cylinder automotive engine. The crankshafts and cam shafts shall have been previously reground, and the surface roughness of all bearing journals and cam surfaces shall be 16 to 20 microinches, (aa). The polishing machine shall then be used to refinish all journals and cam surfaces on the crankshaft and cam shaft. A stop watch shall be used to record the period of contact between the abrasive belt and journal, or cam surface. At the completion of the test, the surface roughness of any bearing journal or cam surface shall be ascertained by use of profilometer. If the surface roughness of any bearing journal or cam surface exceeds eight microinches, aa, or if the average period of contact between the abrasive belt and the bearing journal or cam surface exceeds 30 seconds, the machine shall be rejected (see 3.5.2).

4.6.4 Capacity test. The largest available crankshaft, within the capacity of the machine, shall be mounted between the centers and the bearing journals shall be refinished. Failure to meet the capacity requirements of 3.5.1, or any evidence of malfunction during the refinishing operation, shall be cause for rejection of the machine.

* 4.6.5 Dielectric strength test. The electrical circuits shall be subjected to 1880 volts rms at 60 Hz applied between mutually insulated circuits, between insulated circuits and ground, and between each conductor and the chassis ground. Voltage shall be increased from zero to test voltage at the approximate rate of 500 volts per second, held at test voltage for one minute, and removed at the same rate as applied. During the one minute at test voltage the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge) and breakdown (puncture discharge). Any evidence of flashover, sparkover or breakdown, or damage to the unit due to heat or other effects of the applied voltage, shall be cause for rejection (see 3.7).

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

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5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A or B packing, and military marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The polishing machine covered by this specification is intended for use primarily in producing a smooth finish on crankshaft bearing journals. The polishing machine is also used to refinish camshaft bearing journals and cams. Capacity is sufficient to handle any crankshaft up to, and including, 60 inches in length with bearing journals up to, and including, five inches in diameter with a stroke of at least seven inches.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.7).
(e) Motor operation requirements if different (see 3.6.6.1).
(f) Additional abrasive belts if different (see 3.8).
(g) Painting and finishing if different (see 3.9).
(h) Product identification if different (see 3.10).
(i) First article inspection if different (see 4.3).
(j) Quality conformance inspection if different (see 4.4).
(k) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

* 6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

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* 6.5 Standard lubricants. The polishing machine manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.5) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Rock Island, IL 61299-7300

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-45805C(AL)	Polishing Machine, Automotive Crankshaft and Cam Shaft

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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